

NMAM INSTITUTE OF TECHNOLOGY, NITTE*Off-Campus Centre of Nitte (Deemed to be University)***First / Second Semester B.Tech (CBCS) Degree Examinations**

Summer Semester – July 2024

EC1002-2 – APPLIED DIGITAL LOGIC DESIGN

(For ACT, AD, AM, CB, CC, CS, IS, RI, EE, EC, VLSI)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions each from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. ----- are universal logic gates.
A) NAND and NOR
B) NOR AND EX-OR
C) AND and NOT
D) OR and EX-OR
2. The base of hexadecimal number system is
A) 2
B) 8
C) 10
D) 16
3. Which of the following logic gates provides output as 0 when both inputs are the same (either 0 or 1)?
A) XNOR
B) XOR
C) NOR
D) NAND
4. Which of the following is not a binary number
A) 111
B) 101
C) 11E
D) 000
5. A 3 variable Karnaugh map has _____ cells.
A) 8
B) 3
C) 16
D) 4
6. The Boolean function $A+BC$ is a reduced form of
A) $AB+BC$
B) $(A+B)(A+C)$
C) $A'B+AB'C$
D) $(A+C)B$
7. If a 3-input NOR gate has eight input possibilities, how many of those possibilities will result in a HIGH output?
A) 1
B) 2
C) 7
D) 8
8. If a three variable switching function is expressed as the product of maxterms by $f(A, B, C) = \pi(0, 3, 5, 6)$ then it can also be expressed as sum of min terms by
A) $\pi(1, 2, 4, 7)$
B) $\Sigma(0, 3, 5, 6)$
C) $\Sigma(1, 2, 4, 7)$
D) $\Sigma(1, 2, 3, 7)$
9. Adding 1001 and 0010 gives the output of
A) 1011
B) 1111
C) 1010
D) 0000
10. If A and B are the inputs of a half adder, the sum is given by _____
A) A AND B
B) A OR B
C) A XOR B
D) A EX-NOR B

11. A circuit that converts n inputs to 2^n outputs is called
 A) Encoder B) Decoder
 C) Comparator D) Adder
12. A digital multiplexer is a combinational circuit that selects
 A) One digital information from several sources and transmits the selected one B) Many digital information and convert them into one
 C) Many decimal inputs and transmits the selected information D) Many decimal outputs and accepts the selected information
13. The truth table for an S-R flip-flop has how many VALID entries?
 A) 1 B) 2
 C) 3 D) 4
14. When the J and K inputs are low, the state of the outputs Q and Q' are
 A) Unchanged on clocking B) Changed on clocking
 C) Changed on output D) Changed on input
15. How many select lines would be required for an 8-line-to-1-line multiplexer?
 A) 2 B) 4
 C) 8 D) 3
16. The characteristic equation for T flip flop is
 A) $TQ + T\bar{Q}$ B) $\bar{T}Q + T\bar{Q}$
 C) $\bar{T}Q + \bar{T}\bar{Q}$ D) $T\bar{Q} + T\bar{Q}$
17. The minimum number of flip-flops that can be used to construct a modulus-5 counter is
 A) 3 B) 8
 C) 5 D) 10
18. The group of bits 10110111 is serially shifted (right-most bit first) into an 8-bit parallel output shift register with an initial state 11110000. After two clock pulses, the register contains
 A) 10111000 B) 10110111
 C) 11110000 D) 11111100
19. In which of the following output will always follow a sequence either in downward or upward direction?
 A) Counter B) Registers
 C) Both A and B D) None of these
20. Synchronous counters eliminate the delay problems encountered with asynchronous (ripple) counters because the:
 A) input clock pulses are applied only to the first and last stages B) input clock pulses are applied only to the last stage
 C) input clock pulses are not used to activate any of the counter stages D) input clock pulses are applied simultaneously to each stage

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	Convert the following equations into their requested canonical form				
	(i) $X(A, B, C) = \bar{A}B + BC$ (SOP)				
	(ii) $P(W, X, Y, Z) = (\bar{W} + X)(Y + \bar{Z})$ (POS)				
	(iii) $F(X, Y, Z) = (X + Y)(\bar{X} + Z)$ (POS)	6	L*3	1	1
b)	Design a logic circuit that controls the passage of the signal 'A' according to the following requirement				
	i) Output 'X' will equal 'A' when control inputs B and C are the same				
	ii) 'X' will remain 'HIGH' when B and C are different	6	L4	1	2

- c) Using K-Map, find minimal sum expression for the following switching function.
- $$f(a, b, c, d) = \sum m(1, 3, 4, 5, 7, 8, 9, 11, 15)$$
2. a) Using QM method find the prime implicants of the function.
- $$f(a, b, c) = \sum m(0, 1, 2, 3, 4, 5, 6)$$
- b) Using K-Map, find minimal sum expression for the following switching functions.
- (i) $f(a, b, c) = \sum m(0, 2, 4, 6)$
- (ii) $f(a, b, c) = \sum m(0, 2, 3, 4, 5, 7)$
- (iii) $f(a, b, c) = \sum m(1, 2, 3, 6, 7)$
- c) Simplify the following switching equations using Boolean theorem and realize the minimal expression using basic gates.
- (i) $F = \bar{X}\bar{Y}\bar{Z} + X\bar{Y}Z + \bar{X}Y\bar{Z} + XYZ$
- (ii) $F = (A + \bar{B} + C)(\bar{A} + \bar{B} + C)$
3. a) Perform the code conversion for the following and show each steps clearly.
- (i) $12.6532_{(10)} = (?)_2$
- (ii) $110100111100.0101_{(2)} = (?)_{10}$
- (iii) $CF.52_{(16)} = (?)_{10}$
- b) For the given incompletely specified Boolean function, find the minimal sum.
- $$f(a, b, c, d) = \sum(1, 5, 6, 7, 9, 11, 12, 13) + \sum d(0, 3, 4)$$
- c) State and prove De-Morgan's theorem using the tabular method.
- Unit – II**
4. a) Design half subtractor and full subtractor circuits using logic gates.
- b) Implement $f(a, b, c) = \sum m(1, 4, 5, 7)$ using
- (i) 8:1 MUX
- (ii) 4:1 MUX using a and b as select lines.
- c) Write short note on basic bistable element.
5. a) Explain 2-to-4-line decoder using truth table, output expression, logic diagram. Implement the given function using relevant decoder.
- (i) $F1 = \sum m(0, 1, 3)$
- (ii) $F2 = \sum m(0, 2, 3, 5, 7)$
- b) Explain the operation of master slave JK flip flop with truth table and timing diagram. Write the symbol for the same.
- c) Design 1-bit comparator using basic gates.
6. a) With neat logic diagram and function table explain D Flip flop.
- b) Convert JK Flip flop to
- (i) D Flip flop
- (ii) T Flip flop
- c) Explain 4-bit parallel adder with necessary block diagram.
- Unit – III**
7. a) Explain the working of 3-bit ripple up counter using D flip flops.
- b) What is a shift register? Design a 4-bit shift register using D Flipflops and explain left and right shift operations.
8. a) Design a synchronous counter to sequence $0 \rightarrow 2 \rightarrow 3 \rightarrow 6 \rightarrow 5 \rightarrow 1 \rightarrow 0$ using D flip flops with minimal combinational gating.
- b) What is a Ripple counter? Design and explain 3-bit ripple down counter using T flip flops.

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
